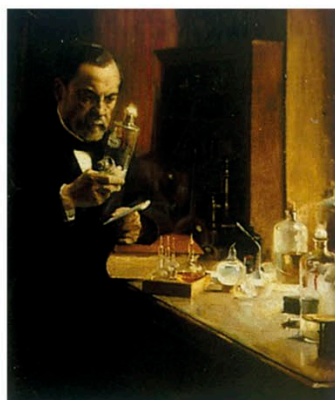


20th Century Onward

For most of us, modern medicine is exemplified by drugs such as antibiotics and highly technical methods of diagnosis and treatment. However, many might be surprised to discover that, for much of the last century, herbal medicines have been the primary form of treatment, even in Western countries.

Even as late as the 1930s, around 90 percent of medicines prescribed by doctors or sold over the counter were herbal in origin. It is only during the last 70 years that laboratory-produced medicines have become the norm. During the First World War (1914–1918), for example, garlic (*Allium sativum*, p. 59) and sphagnum moss (*Sphagnum* spp.) were used by the ton in the battle trenches to dress wounds and to treat infections. Garlic is an excellent natural antibiotic, and was the most effective antiseptic available at the time,



Louis Pasteur was a pioneer in the identification of bacteria.

and sphagnum moss, gathered from the moorlands, makes a natural aseptic dressing.

Science & Medicine

The development of new medicines in the laboratory—either extracted from medicinal plants or synthesized—stretches back to the early 19th century, when chemists first isolated constituents such as morphine, from opium poppy (*Papaver somniferum*, p. 244), and cocaine,

from coca (*Erythroxylum coca*, p. 206). From that time onward, scientists made tremendous progress in understanding how isolated chemicals affect the body, as well as how the body works in health and disease. From the 1860s, scientists—most notably Louis Pasteur (1822–1895)—began to identify the microorganisms that were ultimately responsible for causing infectious diseases, such as tuberculosis and malaria.

Naturally enough, the first aim of those engaged in medical research was to seek out medicines that would act as “magic bullets,” directly attacking the microorganisms concerned and ridding the body of the threat. This eventually led to the discovery, or, more accurately, the rediscovery of penicillin by a number of medical researchers, most notably Alexander Fleming (1881–1955) in 1929. However, while 20th-century scientists were the first to scientifically evaluate antibiotics as medicines, they were not the first to employ them in healing. Antibiotic molds had been grown and used to combat infection in ancient Egypt, 14th-century Peru, and in recent European folk medicine.

In the decades following the Second World War (1939–1945), when antibiotics first came into use, it seemed as though a new era had dawned in which infection could

be conquered, and life-threatening diseases such as syphilis, pneumonia, and tuberculosis would cease to be major causes of death in the developed world. Modern medicine also provided other highly effective drugs such as steroid anti-inflammatories, and it seemed as if it was simply a question of time until cures for most illnesses were found.

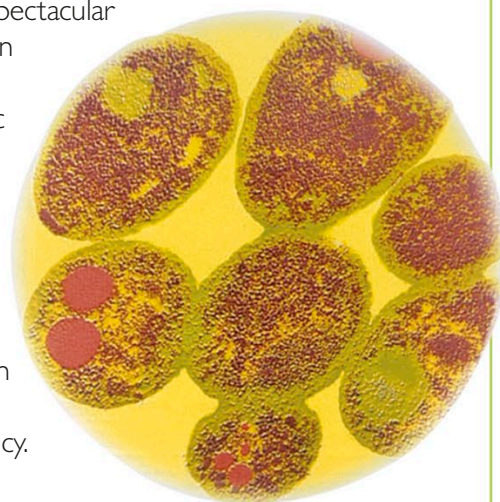
Ascendancy of Biomedicine

As Americans and Europeans became accustomed to medication that led to an almost instant short-term improvement in symptoms (if not in underlying health), herbal medicines came to be seen by the public as outmoded and ineffective. Increasingly, the practice of herbal medicine was outlawed in North America and most of Europe, while the wealthy in developing countries abandoned herbal medicine in favor of the new treatments available.

This was in no small part due to the medical profession itself, which saw herbal medicine as a throwback to the superstitions of the past. From the late 19th century onward, the aim of organizations such as the American Medical Association and the British Medical Association had been to monopolize conventional medical practice. Herbal medicine thus neared extinction in many countries, especially in the U.S. and Britain. In Britain, for example, from 1941 until 1968 it was illegal to practice herbal medicine without medical qualifications.

The Tide Turns

Although there were spectacular successes with modern chemical medicines, there were also horrific disasters, most notably the thalidomide tragedy in 1962 in Britain and Germany, when 3,000 deformed babies were born to mothers who had taken the drug for morning sickness during pregnancy. This event marked a turning point in the public's opinion of chemical medicines.



An electron micrograph of the parasite Plasmodium. Transmitted to man by infected mosquitoes, it causes the disease malaria.

